Night Drive? Not Yet. Investigating Performance Drops in Autonomous Driving Models at Night.

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Abstract

Light Detection and Ranging (LiDAR) technology plays a crucial role in autonomous driving systems, providing precise distance measurements and depth perception. However, LiDAR-based object detection systems experience performance degradation under low-light conditions, such as nighttime driving, which is surprising given that LiDAR operates independently of visible light. Our research confirms and extends previous findings, demonstrating a significant drop in mean Average Precision (mAP) for LiDAR models during nighttime using the NuScenes dataset. We investigate the underlying causes, including data imbalance and differences in the number of ground truth bounding boxes between day and night scenes. Our experiments include limiting bounding box predictions and retraining models on balanced datasets. Results show that retraining on balanced datasets improves nighttime performance, though challenges remain for certain object classes. We conclude that while balancing data helps, further research is needed to enhance the robustness of LiDAR-based object detection in varying lighting conditions. Potential solutions include increasing night scene data, employing advanced augmentation techniques, and exploring domain adaptation methods.

1. Introduction

Light Detection and Ranging (LiDAR) technology has become a pillar in the development of autonomous driving systems. As its core, LiDAR uses lasers mounted to a vehicle to create a detailed point-cloud representation of the surrounding environment. However, despite its advantages in providing precise distance measurements and depth perception, LiDAR-based object detection systems show performance degradation under low-light conditions, such as when driving during nighttime. This phenomenon is surprising, given that LiDAR sensors, relying on the reflection of laser pulses, operate independently of visible light.

Recent studies have consistently shown that LiDAR-based detection models, both LiDAR-only and those that combine LiDAR with camera data, perform worse at night compared to daytime. For instance, Cai et al. [3] demonstrated a substantial drop in mean Average Precision (mAP) when evaluating LiDAR models on night scenes using the NuScenes dataset [2]. Our research corroborates these findings, showing similar performance declines across multiple LiDAR detection models when assessed under nighttime conditions.

This performance discrepancy has far-reaching implications for the deployment of autonomous vehicles, as reliable object detection is crucial for safe navigation at all times of the day. Accurate detection and classification of other road users, such as vehicles and pedestrians, are paramount to avoid collisions and ensure smooth traffic. The degradation in detection performance at night therefore poses a significant challenge, compromising the performance, and potentially, safety of autonomous systems. Our study delves into the underlying causes of this issue, examining factors such as differences in the number of ground truth bounding boxes during day versus night, and data imbalance, meaning a significant difference in total day versus night scenes. Furthermore, we explore potential solutions to mitigate the performance gap between day and night detection, aiming to enhance the robustness and reliability of LiDAR-based object detection in all lighting conditions.

1.1. State of the Art in LiDAR-Based Object Detection

LiDAR-based object detection has advanced significantly in recent years, driven by the growing demands of autonomous driving and technical conditions to realise it. State-of-the-art methods can be broadly categorized into three main approaches: LiDAR-only detection, multi-modal fusion of Li-

DAR and camera data, and domain adaptation techniques.

LiDAR-Only Detection. LiDAR-only methods utilize the point clouds generated by LiDAR sensors to detect and classify objects in 3D space. Notable works include VoxelNet [15], which voxelizes point clouds and applies 3D convolutions, and PointPillars [6], which structures point clouds into pillars for efficient processing. These methods have demonstrated robust performance in clear conditions but often struggle with sparse data points for small or distant objects.

Multi-Modal Fusion. To overcome the limitations of LiDAR-only systems, multi-modal approaches that combine LiDAR data with camera images have been developed. These methods leverage the complementary strengths of both sensors, with cameras providing rich color and texture information and LiDAR offering precise depth data. Prominent examples include PointPainting [11], which fuses LiDAR points with semantic segmentation from images, and BEVFusion [9], which integrates bird's eye view representations of both modalities. Despite their advancements, these methods can suffer from misalignment issues and degraded performance under poor illumination conditions, as noted in TransFusion [1], which introduces a soft-association mechanism to enhance robustness against such challenges.

Domain Adaptation. Addressing domain shifts, such as changes in geographic locations or lighting conditions, is critical for the generalization of detection models. Techniques like Semantic Point Generation (SPG) [13] and cross-modal unsupervised domain adaptation (xMUDA) [4] aim to improve model performance across diverse domains by augmenting point clouds and leveraging multi-modal learning. These approaches have shown effectiveness in mitigating performance drops due to domain shifts, although they often require access to target domain data during training, which is not always feasible. For example, Wang et al. [12] investigated adapting 3D object detectors trained in Germany to test in the USA, identifying car size differences as a major adaptation hurdle and proposing a simple correction based on average car size to significantly improve cross-dataset performance.

1.2. Our Research Focus

In this paper, we investigate the specific challenges posed by nighttime environments for LiDAR-based object detection. Our study is structured around three main research issues:

Confirmation of the Issue. We replicate and extend the findings of previous research by evaluating several LiDAR detection models on the NuScenes dataset, which contains both day and night scenes. We test if these models consistently perform worse at night, regardless of whether they use LiDAR alone or in combination with camera data.

Dataset Exploration. We conduct a detailed analysis of the NuScenes dataset to identify significant imbalances between day and night scenes. This involves examining the distribution of scenes and bounding boxes, as well as the class distributions, to understand how these factors might influence model performance.

Hypothesis Testing and Model Retraining. Lastly, we explore two hypotheses to understand the root causes of the performance drop at night. The first hypothesis is that models predict too many bounding boxes in night scenes, leading to empty detections, which will be tested by limiting the number of bounding boxes during evaluation. The second hypothesis addresses data imbalance, suggesting that the performance drop at night is due to the fewer night scenes available during training compared to day scenes. We test this second hypothesis by retraining models on balanced datasets with equal proportions of day and night scenes.

2. Related Work

The recent advancements in 3D object detection and semantic segmentation have brought significant progress to autonomous driving and other applications requiring 3D scene understanding. However, several challenges persist, particularly in cross-modal unsupervised domain adaptation (UDA), multi-modal fusion techniques, and test-time adaptation strategies. This section reviews relevant works in these areas, highlighting the evolution and effectiveness of various approaches.

2.1. Cross-Modal Unsupervised Domain Adaptation (UDA)

Cross-modal UDA aims to bridge the performance gap between different domains by leveraging the complementary strengths of various sensor modalities. Liu et al. [7] introduced a cross-modal UDA framework for 3D semantic segmentation, which addresses the heterogeneity of input spaces, such as 2D images and 3D point clouds, by enabling mutual learning between modalities. Similarly, Jaritz et al. [4] proposed xMUDA, where modalities learn from each other through mutual mimicking, disentangled from the segmentation objective. This prevents the stronger modality from adopting false predictions from the weaker one, significantly improving UDA performance across various scenarios.

2.2. Multi-Modal Fusion Techniques

Effective multi-modal fusion is crucial for enhancing 3D object detection performance. Caesar et al. [2] introduced the NuScenes autonomous driving dataset, facilitating robust multi-modal fusion by providing comprehensive LiDAR and camera data. This dataset has been instrumental in developing methods that integrate data from multiple sensors for improved detection accuracy. Zhou and Tuzel [15] proposed VoxelNet, an end-to-end learning framework that utilizes voxel-based representations for 3D object detection from point clouds, demonstrating the potential of voxel features in capturing geometric information. Building on this, Lang et al. [6] developed PointPillars, which organizes point cloud data into vertical columns (pillars) and employs a PointNet to learn features for each pillar, enhancing detection efficiency and accuracy.

Bai et al. [1] introduced TransFusion, a robust LiDAR-camera fusion method using transformers. This approach aligns features from different modalities at the object level, ensuring precise spatial alignment without requiring camera-to-BEV transformations, thereby improving detection performance. Similarly, the ObjectFusion framework by Cai et al. [3] further enhances multi-modal fusion by eliminating inter-modality transformations during fusion, preserving the primary features of each modality and enabling precise object-level fusion.

2.3. Test-Time Adaptation (TTA) Strategies

Test-time adaptation (TTA) has emerged as a practical approach to handle domain shifts during deployment. Liu et al. [8] proposed Night-TTA, a TTA framework designed for nighttime color-thermal semantic segmentation. Night-TTA addresses the significant domain gap between daytime and nighttime data by employing imaging heterogeneity refinement and class-aware refinement strategies. This approach ensures reliable performance by leveraging cross-modal invariant features and robust ensemble learning during testing.

Similarly, Xu et al. [13] introduced Semantic Point Generation (SPG), targeting UDA for 3D object detection. SPG generates semantic points to recover missing parts of foreground objects in point clouds, thereby enhancing the reliability of LiDAR detectors under domain shifts. This method significantly improves detection performance across different geographic locations and weather conditions, demonstrating its effectiveness in real-world scenarios.

Building on these advancements, our study focuses on the specific challenges posed by nighttime environments for LiDAR-based object detection. We leverage the NuScenes dataset [2], which provides comprehensive day and night scenes, to investigate performance discrepancies.

Table 1. Number day versus night scenes in the NuScenes dataset.

Data Split	Day	Night
Training Set	616	84
Validation Set	135	15
Total	751	99

3. Dataset Analysis

3.1. NuScenes

NuScenes is a multimodal dataset collected and developed for autonomous driving. It was the first dataset to carry 6 cameras, 5 radars and 1 LiDAR sensor. The scenes are collected in Boston and Singapore, in various weather conditions ranging from clear weather through nighttime and rain. It is also the first multimodal dataset to contain object attributes and scene descriptions on top of object class and localization in nighttime and rainy conditions [2]. The focus of this paper is on the night scenes, which make up 11.6% of the data.

3.2. Dataset Statistics.

In this section, we analyze the NuScenes dataset, finding significant imbalances between day and night scenes which informed our subsequent experiments on the detection performance drop in night scenes.

Scene Distribution.
The NuScenes dataset is divided into training and validation sets with a notable disparity in the number of day and night scenes summarized in Table 1.

We find that 88.5% of all scenes included in our study are day scenes, making the dataset highly imbalanced.

Bounding Box Distributions.
To understand the effect of the day-night imbalance, we conducted a detailed statistical analysis of the bounding box distributions for various object categories in both day and night scenes.

Normality Test. We first assessed the normality of the data distributions using the Shapiro-Wilk test. The results indicated that the data does not follow a normal distribution (all $p < 0.05$). This necessitated the use of non-parametric tests for analysis.

Mann-Whitney U Test. We employed the Mann-Whitney U test to compare the means of day and night scenes for each category of vehicles and pedestrians. This test is suitable for comparing distributions that do not meet the assumptions of normality. The Mann-Whitney U test evaluates whether there is a significant difference between the two independent samples.

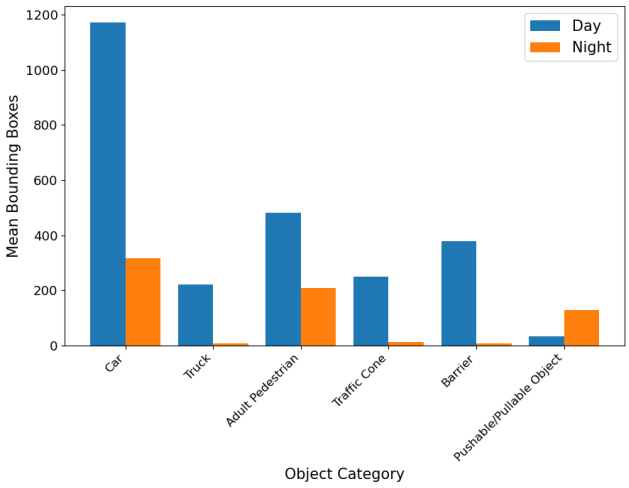


Figure 1. Mean counts for selected categories during day and night of the training set. For example, an average day scene included 1172 car detections, while an average night scene included only 316 car detections. Notably, pushable objects are the only class observed more often at night.

Bonferroni Correction. To account for the multiple comparisons problem, we applied the Bonferroni correction to the significance level ($\alpha = 0.05$). The multiple comparisons problem increases the risk of Type I errors (false positives) due to the number of comparisons being made. Therefore, the original p-value threshold of 0.05 is divided by the number of tests conducted to maintain the overall error rate. Given the number of tests, the adjusted threshold for significance was set at 0.00217.

Results of the Analysis. For the training set analysis reveals significant differences in the number of bounding boxes between day and night scenes for various object categories. Notably, 10 out of 23 categories such as car, truck, adult pedestrian, trafficcone, and barrier have significantly higher counts during the day. Conversely, a single category - pushable/pullable objects - showed a significantly higher count at night.

In the validation set, similar trends were observed. The car, truck, and adult pedestrian categories again had significantly more bounding boxes during the day, despite the smaller amount of scenes which made detecting a significant effect more challenging. Additionally, the pushable/pullable objects category continued to show higher counts at night.

Figure 1 shows an graphical overview of the number of object bounding box counts per category. The full results from our statistical analysis can be found in Appendix A.

Moving beyond object categories, the average number of total bounding boxes per scene is 2780.09 for day scenes and 737.32 for night scenes. Consequently, day scenes contain approximately 3.77 times more bounding boxes than night scenes.

Table 2. Results of reproducing results from [3]. We report mAP(%). In the 'Modality' column, L denotes LiDAR-only models and LC denotes models using both LiDAR and camera data.

Model	Modality	Theirs		Ours	
		Day	Night	Day	Night
CenterPoint	L	62.8	35.4	59.6	33.9
BevFusion	LC	68.5	42.8	67.8	42.2
TransFusion-L	L	-	-	64.7	37.5

In conclusion, the statistical analysis reveals the following points:

1. There are significantly fewer night scenes than day scenes in both the training and validation sets.
2. Night scenes have significantly fewer bounding boxes compared to day scenes.
3. The class balance is different for day and night scenes across most object categories.

4. Experiments

4.1. Reproduction

In order to perform further experiments, we first reproduce the results from Cai et al. [3], showing that there is a difference in performance on day and night scenes. To do so, we evaluated pre-trained models from OpenPCDet [10] on the NuScenes validation set. Results can be found in Table 2. We report mean Average Precision (mAP), following Cai et al. [3]. We were able to reproduce the results and see that the models perform drastically worse on night scenes than day scenes. Note that this occurs for both LiDAR-only models and models using both LiDAR and camera data.

4.2. Limiting bounding boxes

We have seen in section 3 that there is a significant difference in the number of bounding boxes between day and night scenes. Following this, we hypothesize that the model has learned to predict many bounding boxes for each scene. For day scenes, this would lead to a correct number of bounding boxes, but for night scenes this could lead to many empty boxes, degrading the performance. To test this hypothesis, we limit the number of bounding boxes counted in evaluation. Results can be seen in Figure 2. We see that the night performance stays mostly stable as the number of predicted bounding boxes increases. Furthermore, we see that the day performance increases as the number of bounding boxes increases. This indicates that the model predicts the correct number of bounding boxes for both day and night scenes.

Table 3. Comparison of two retrained CenterPoint models. CP-balanced is trained on a subset of the original dataset containing 50% night and 50% day data. CP-unbalanced is trained on a subset of similar size containing 88% day and 12% night data. Both models were trained from scratch with the same hyperparameters. CP-balanced achieves the same performance on day data, while improving on night data, showing the importance of data balance.

Setting	Model	Class (AP)										mAP
		Car	Truck	Bus	Trailer	Constr.	Pedest.	Motorc.	Bicycle	T. Cone	Barrier	
Day	CP-Balanced	0.824	0.517	**0.653**	0.283	**0.162**	0.829	**0.470**	0.316	0.633	0.618	**0.531**
	CP-Unbalanced	**0.828**	**0.533**	0.634	**0.287**	0.118	**0.832**	0.441	**0.324**	**0.655**	**0.654**	0.531
Night	CP-Balanced	**0.845**	**0.632**	0.000	0.000	0.000	0.464	**0.685**	**0.172**	0.000	0.266	**0.307**
	CP-Unbalanced	0.836	0.564	0.000	0.000	0.000	**0.495**	0.623	0.125	0.000	**0.280**	0.292

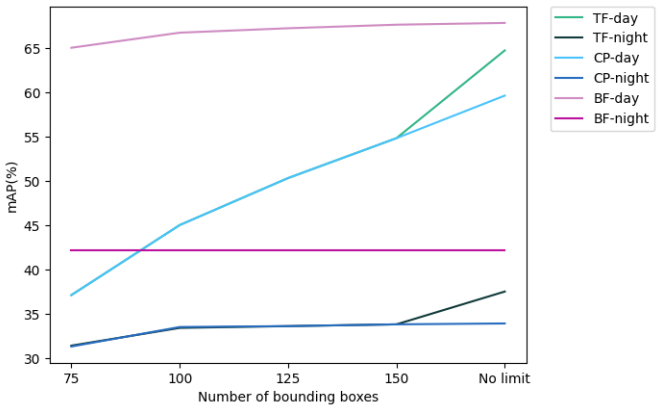


Figure 2. Results of limiting the number of predicted bounding boxes in evaluation. TF denotes TransFusion-L, CP denotes CenterPoint and BF denotes BevFusion.

4.3. Retraining models

Another cause of the difference in performance on day and night scenes could simply be data imbalance. As seen in section 3 there is much less data available from night scenes than day scenes. The model could simply be focusing on optimizing performance on day data. To test this, we retrained the CenterPoint model with two settings. One was trained on a balanced dataset, containing 50% day and 50% night data. We refer to this model as CP-balanced. The other was trained on the same amount of data as CP-balanced, but with 88% day and 11% night data, as in the original NuScenes dataset. Since the models were trained on the same amount of data, we can fairly compare them. We will refer to this model as CP-unbalanced.

We used a total of 685 scenes for retraining both models. Because OpenPCDet did not have a built-in way to separate the day and night scenes, we had to perform this separation using the xMUDA repository [5]. Both models were retrained from scratch for 20 epochs, using the same set of hyperparameters, following the original setting [14]. For full training details and hyperparameters, see Appendix B. We ran evaluation on the day and night data from the validation set. The results can be seen in Table 3.

Most importantly, we see that CP-balanced achieves a higher mAP on the night data than CP-unbalanced. We see a significant improvement on the truck class at night especially. It even improves on the car, which is the most represented in the data. Additionally, CP-balanced manages to reach the same mAP on the day data, even though it is trained on less day data than CP-unbalanced. This indicates the importance of training on a balanced dataset. However, we still see a large gap in performance on day and night data for some classes, like Pedestrian or Bicycle. Interestingly, there are some classes where performance is higher for night data than day data, indicating possible relations between those classes being more frequent at night.

While these results reaffirm our notion on data imbalance, there is still a large gap between day and night performance, even when the model is trained on a balanced dataset. For both CP-balanced and CP-unbalanced, this is a direct result of the model failing to get any correct prediction on the bus, trailer, construction vehicle and traffic cone classes at night. The AP of zero on these classes drags down night mAP significantly.

4.4. Investigating class imbalance

In order to understand why the models fail to get any correct predictions on these classes, we consider the amount of instances for each class in both day and night data, as can be seen in Figure 3. For the classes bus, trailer and construction vehicle, we can see in Figure 3 that there are 0 instances in the night scenes of the validation set. As a result, the models cannot score higher than 0% AP on these classes. Therefore, we recalculate the AP without these classes in Table 4. Excluding zero-instance classes increases mAP with 0.131% mAP for the CP-balanced model at the night setting, which closes the gap between night and day performance a bit more.

In addition to bus, trailer and construction vehicle, both models score 0% AP on the traffic cone class. This is not an empty class, however, meaning we cannot remove this from the calculation. A possible explanation for this score

Table 4. Recalculated mAP for CP-balanced and CP-unbalanced. The classes bus, trailer and construction vehicle are excluded from the calculation as there are 0 instances in the night scenes of the validation set.

Setting	Model	mAP
Day	CP-balanced	0.601
	CP-unbalanced	0.615
Night	CP-balanced	0.438
	CP-unbalanced	0.418

can be the large gap between instances in day scenes and night scenes, as we can see from Figure 3. Further research on why the models fail to detect traffic cones would be beneficial.

5. Discussion

Our study confirmed the significant drop in mean Average Precision (mAP) for LiDAR-based object detection systems during nighttime conditions, consistent with previous research. Several key factors contributing to this performance gap were identified.

5.1. Data Imbalance

The primary issue identified was the imbalance in the number of day versus night scenes in the NuScenes dataset. With 88.5% of scenes being day scenes, models had more exposure to day data, skewing their performance. Retraining models on balanced datasets improved night performance, as seen with the CP-balanced model achieving higher mAP on night data. However, despite the improvements, there remains a noticeable gap in performance between day and night scenes for certain classes, such as Pedestrian and Bicycle. This suggests that while data balancing is beneficial, it is not a complete solution.

5.2. Class Imbalance

The investigation into class-specific performance revealed that the absence of certain object classes in the night scenes of the validation set contributes to the models' poor performance. For example, the classes bus, trailer, and construction vehicle had no instances in night scenes, leading to an inevitable AP of zero for these classes. When these zero-instance classes were excluded from the calculation, the mAP for night scenes improved slightly, but not enough to bridge the overall performance gap.

Additionally, the traffic cone class, which was not an empty class, still scored 0% AP at night. This could be due to the significantly lower number of instances in night scenes compared to day scenes, indicating that even within classes present in both day and night scenes, there is a substantial imbalance which affects detection accuracy.

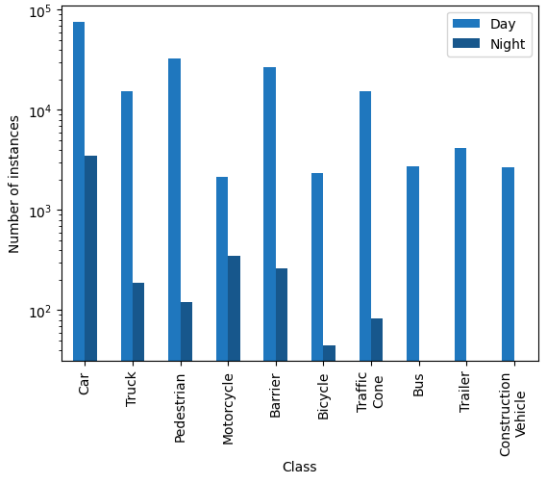


Figure 3. Number of instances per class in NuScenes validation set.

5.3. Bounding Box Predictions

Another test case was the effect of bounding box predictions on performance. Limiting the number of bounding boxes during evaluation showed that the model predicts an appropriate number of bounding boxes for both day and night scenes. This suggests that the issue lies not in the number of predicted boxes but potentially in their quality and accuracy under different lighting conditions.

5.4. Future Work

Future work could focus on several areas to enhance the performance of LiDAR-based object detection systems in night conditions. Firstly, increasing the night scene data and ensuring a balanced distribution of object classes could be worth looking into. Additionally, implementing advanced augmentation techniques such as generating synthetic night data can help mitigate the data imbalance. Lastly, exploring domain adaptation methods could be essential for enabling models to perform reliably across varying lighting conditions.

6. Conclusion

NuScenes dataset is highly unbalanced from night and day standpoint. First explored strategy involved limiting the amount of predicted bounding boxes per scene has not proven to be an effective way improve the performance. Second strategy of retraining on a balanced dataset, was able to achieve small improvement. While the gap between performances was not closed, an insight possibly explain-

ing the difference was obtained. The analysis shows that the types of classes differ heavily between day and night. The absence of these classes impacts the overall performance metrics, skewing the results and highlighting the class imbalance issue. It is important to focus on developing specialized models that are able to better handle the class distribution of the night scenes.

7. Process

7.1. Difficulties and Limitations

The most difficult thing we encountered during this project was getting the OpenPCDet codebase to run. Package mismatches and conflicting CUDA versions meant that this took us the better part of three weeks to get up and running. During this time, we were able to perform the data analysis on NuScenes and some literature analysis, but we would have rather had extra training time. Another great difficulty lie in the separating of the dataset in day and night data. Although NuScenes does have the information of day/night scenes, it is stored in a .json file in the metadata. OpenPCDet did not have an in-built way of separating the dataset into day and night or evaluating on them separately. Finding how to separate the data into day and night ánd adapt the OpenPCDet code to accept a subset of the dataset was our second major hurdle. Finally, we had issues with managing the data. The NuScenes dataset is very big and we had to redownload it a couple times, due to it being deleted from Snellius, or Snellius being down. Each time, this took a lot of work.

7.2. Division of Work

Lennard: Set up data and shared cloud environment. Run evaluation using pre-trained models.
Lisann: Set up project environment in Snellius. Statistical analysis of the dataset (see section 3.2). Co-created poster (with Livia and Marga). Wrote sections 1 (all), 2 (all), and 3.2 of the report.
Livia: Methodology, conclusion, poster co-creation, literature research, data splitting.
Marga: Bounding box analysis (section 4.2), poster co-creation, writing of sections 4.1 and 4.2, organizational issues (planning meetings, keeping meeting notes and keeping track of to-do's)
Bart: Run evaluation of both pre-trained and retrained models on day/night data. Set up and run retraining for CP-balanced and CP-unbalanced. Writing of Retraining models (section 4.3) and Investigating class imbalance (Section 4.4).
Robbert: separated scenes into day and night. Wrote code for bounding box analysis. Wrote discussion and abstract.

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A. Detailed Statistical Analysis of Bounding Box Distributions per Object Category

The results from our analysis are summarized in Tables 5 and 6 for the training and validation sets, respectively. The tables provide the mean and standard deviation of bounding box counts for day and night scenes, along with the Mann-Whitney U test statistics, and significance levels.

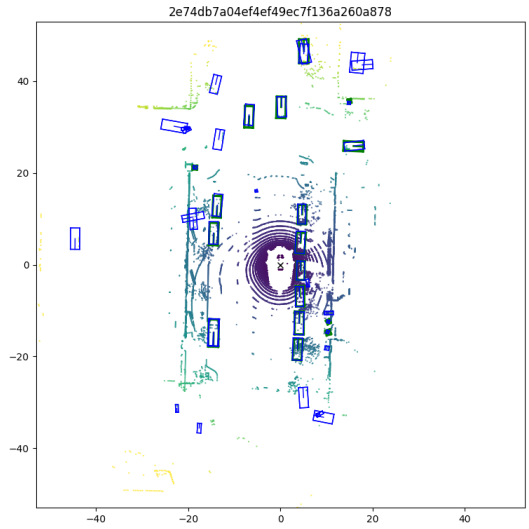


Figure 4. LiDAR point detection of a typical day scene with a variety of detected objects.

B. Retraining Details

Table 7 contains the full training details and hyper-parameters used to retrain both CP-balanced and CP-unbalanced. The configuration was taken directly from the `cbgs_voxel0075_res3d_centerpoint.yaml` from the OpenPCDet repository. We only changed the data the model is trained on.

Table 5. Comparison of bounding box counts of day and night scenes (train set), including standard deviations and significance levels. Large standard deviations reflect the substantial difference in bounding boxes scene-to-scene. Bonferroni-adjusted p-threshold for 5% significance: 0.217%, with 11 out of 23 categories being significant. 'Pushable pullable' objects is the only class more frequent at night, while 10 categories were more frequent at day. *Note:* Non-significance often occurs in rare categories.

Object Category	Mean (day)	Std. (day)	Mean (night)	Std. (night)	p-value	Significant
vehicle.bus.bendy	4.29	22.49	0.64	4.51	0.31	False
vehicle.motorcycle	25.50	55.52	17.62	23.09	0.12	False
movable_object.pushable_pullable	33.16	165.54	128.99	259.58	0.00	more at night
human.pedestrian.stroller	2.95	15.40	0.18	1.63	0.10	False
movable_object.debris	6.78	26.86	0.00	0.00	0.00	False
vehicle.car	1171.76	974.93	315.90	219.22	0.00	more at day
human.pedestrian.child	6.10	32.89	0.07	0.65	0.03	False
vehicle.truck	221.43	239.61	8.74	19.93	0.00	more at day
vehicle.emergency.ambulance	0.06	1.53	0.00	0.00	0.71	False
human.pedestrian.adult	482.57	695.71	208.35	295.64	0.00	more at day
human.pedestrian.wheelchair	1.63	10.85	0.00	0.00	0.12	False
human.pedestrian.personal_mobility	1.07	8.43	0.48	2.93	0.49	False
vehicle.construction	37.69	76.01	0.87	4.27	0.00	more at day
vehicle.bus.rigid	28.42	49.60	28.23	34.11	0.08	False
static_object.bicycle_rack	7.15	20.48	0.00	0.00	0.00	more at day
movable_object.trafficcone	249.88	504.86	12.92	29.14	0.00	more at day
human.pedestrian.construction_worker	24.20	74.14	0.00	0.00	0.00	more at day
vehicle.trailer	64.35	163.98	0.48	4.34	0.00	more at day
vehicle.emergency.police	1.67	11.72	0.60	3.92	0.82	False
animal	1.97	13.39	0.70	4.06	0.62	False
movable_object.barrier	377.55	758.11	7.35	26.83	0.00	more at day
human.pedestrian.police_officer	2.05	13.24	0.00	0.00	0.07	False
vehicle.bicycle	27.85	49.02	5.23	15.34	0.00	more at day

Table 6. Comparison of bounding box counts of day and night scenes (validation set), including standard deviations and significance levels. The Bonferroni-adjusted p-threshold for 5% significance is 0.217%.

Object Category	Mean (day)	Std. (day)	Mean (night)	Std. (night)	p-value	Significant
vehicle.bus.bendy	5.93	23.37	0.00	0.00	0.256	False
vehicle.motorcycle	23.71	43.13	23.07	28.29	0.551	False
movable_object.pushable_pullable	13.73	32.64	150.47	215.86	0.000	more at night
human.pedestrian.stroller	1.94	10.60	0.00	0.00	0.373	False
movable_object.debris	9.03	34.14	0.00	0.00	0.195	False
vehicle.car	989.68	675.80	230.47	140.75	0.000	more at day
human.pedestrian.child	2.59	16.10	0.00	0.00	0.412	False
vehicle.truck	206.83	238.23	12.60	20.83	0.000	more at day
vehicle.emergency.ambulance	0.44	5.14	0.00	0.00	0.756	False
human.pedestrian.adult	438.38	567.81	8.00	14.93	0.000	more at day
human.pedestrian.wheelchair	0.00	0.00	0.00	0.00	1.000	False
human.pedestrian.personal_mobility	0.24	2.74	0.00	0.00	0.756	False
vehicle.construction	35.19	59.26	0.00	0.00	0.003	False
vehicle.bus.rigid	33.56	49.14	0.00	0.00	0.001	more at day
static_object.bicycle_rack	3.44	12.31	0.00	0.00	0.213	False
movable_object.trafficcone	208.16	369.40	5.47	11.61	0.000	more at day
human.pedestrian.construction_worker	17.85	56.33	0.00	0.00	0.078	False
vehicle.trailer	58.57	163.90	0.00	0.00	0.010	False
vehicle.emergency.police	1.08	8.84	0.00	0.00	0.648	False
animal	0.55	4.93	0.00	0.00	0.648	False
movable_object.barrier	362.95	648.23	17.27	64.61	0.003	False
human.pedestrian.police_officer	1.10	8.79	0.00	0.00	0.570	False
vehicle.bicycle	31.98	47.14	3.00	5.30	0.009	False

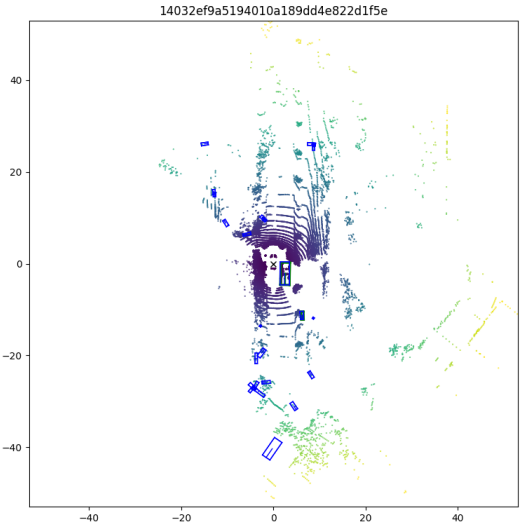


Figure 5. Lidar point detection of a night scene in which there are only a few vehicles and many objects of minor classes (e.g. pushable objects) detected.

Table 7. Full training details and hyperparameters for retraining both CP-balanced and CP-unbalanced. Taken directly from this configuration file.

Parameter	Value
Epochs	20
Batch size	4
Optimizer	adam_onecycle
Learning rate	0.001
Weight decay	0.01
Momentum	0.9
Decay step list	[35,45]
Learning rate decay	0.1
Learning rate clip	$1 * 10^{-7}$
Learning rate warmup	False
Class loss weight	1.0
Location loss weight	0.25
Code loss weights	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 0.2, 0.2, 1.0, 1.0]

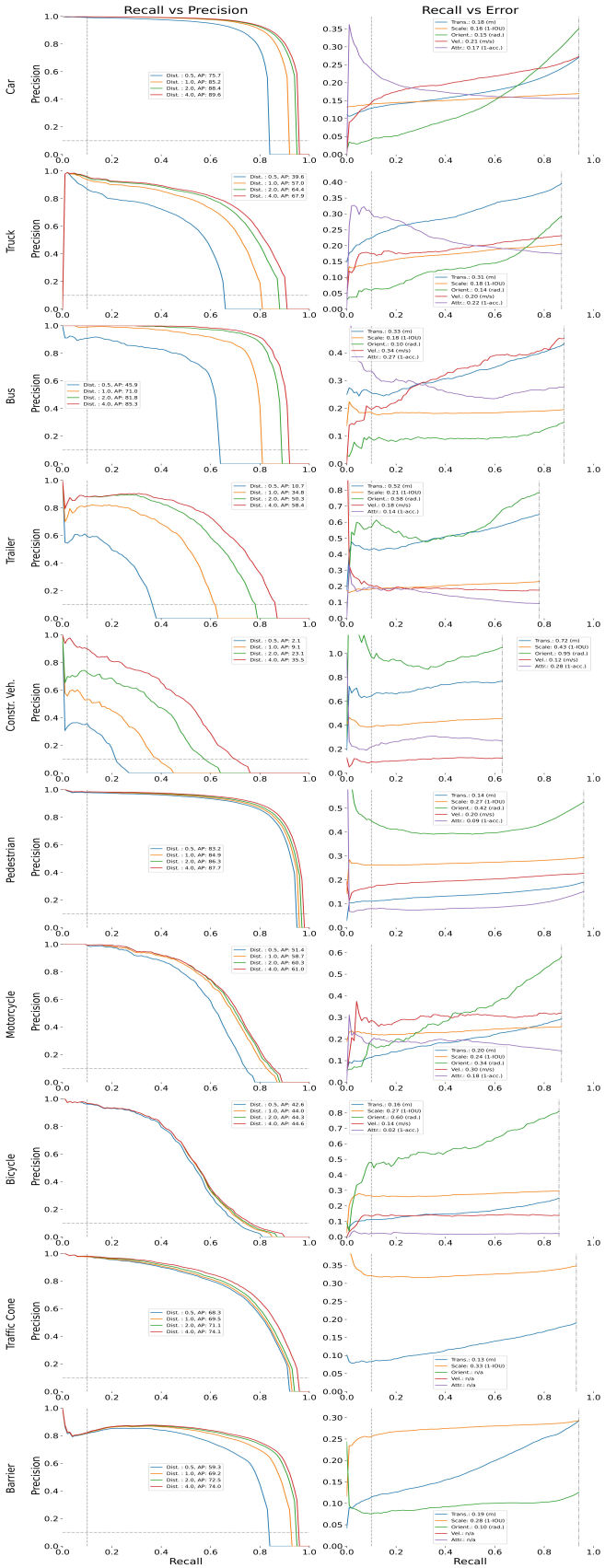


Figure 6. Detailed results of pre-trained Centerpoint model on day scenes in the validation set, split per class.

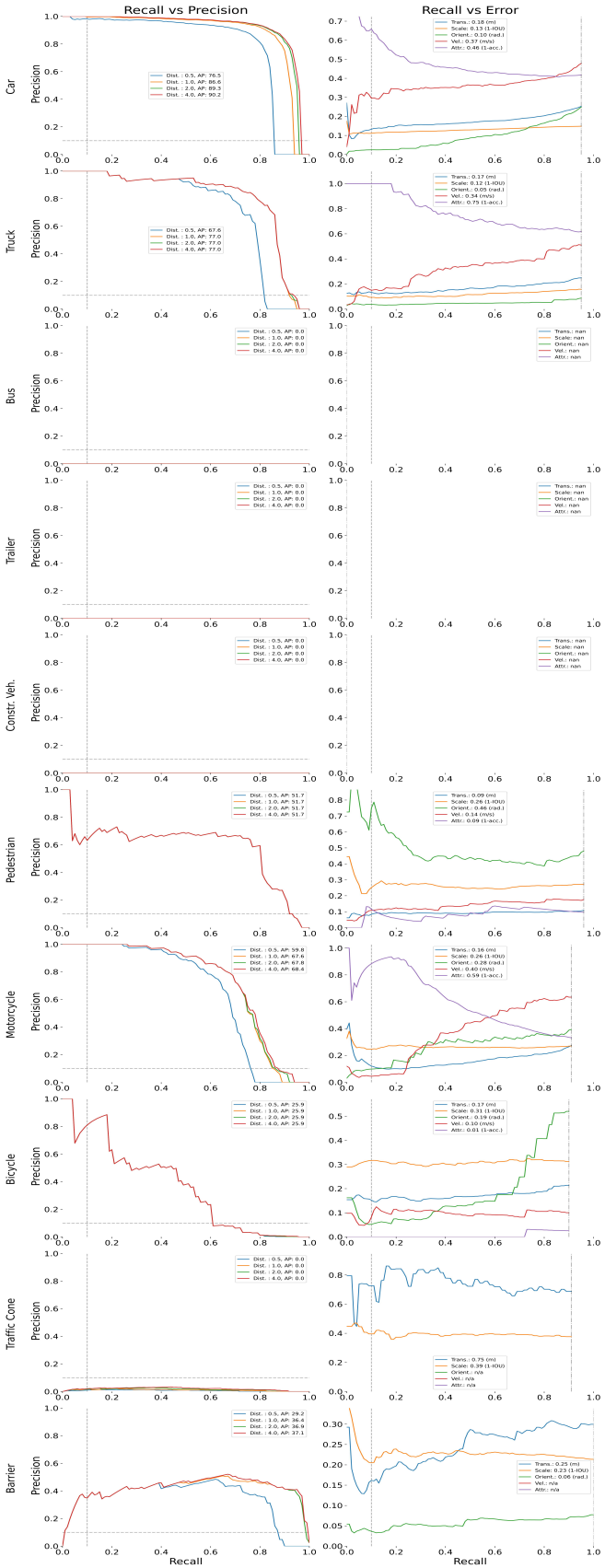


Figure 7. Detailed results of pre-trained Centerpoint model on night scenes in the validation set, split per class.